Junxian Ma

+86 177 1058 6602 | junxianma93@gmail.com | edgep.me | github.com/EdgePro001

RESEARCH INTERESTS

Focus: LLM inference systems, sparse attention, GPU kernel optimization, and algorithm-system co-design

EDUCATION

Tsinghua University *2024 - 2028 (Expected)*
B.Eng. Candidate in Computer Science and Technology | Beijing, China

SELECTED RESEARCH AND SYSTEMS PROJECTS

FusedIndexTopK | Indexer + Top-K Fusion for DeepSeek DSA | GitHub *Jul 2026 - Present*
Independent GPU Systems Research | DeepSeek-V3.2 DSA | NVIDIA H20

- Designed and implemented an exact fused CUDA operator for the DSA discovery stage, eliminating the dense FP32 score write/read between the Indexer and Top-K.
- Co-designed DeepGEMM-style FP8 score production with compact candidate generation, radix selection, and bounded-memory exact repair.
- Reduced end-to-end Indexer+Top-K latency in all 40 held-out H20 cases: median 14.7% versus DeepGEMM+FlashInfer (95% CI: 13.2-15.9%) and 30.1% versus DeepGEMM+torch.topk.
- Extended exact selection to 32K-128K with fixed-memory chunking and online Top-2,048 merging, retaining 4.9-7.8% gains; open-sourced code, tests, and benchmark artifacts.

Sparse-Attention Algorithm-Infra Co-design | NSA-Exp *2026*
Independent Systems Analysis and H100 Experiments
Writing: Sparse Attention: Algorithm-Infra Co-design

- Compared NSA and DeepSeek-V3.2 DSA from token selection to GPU execution, identifying how block layout, head sharing, and memory movement determine whether theoretical sparsity becomes real speedup.
- Profiled the public DSA pipeline from 32K to 2M context on H100. Discovery (Indexer+Top-K) already exceeded FlashMLA at the 64K checkpoint and widened rapidly with context length; eight randomized row-order tests stayed within 0.47% of sorted access, indicating that finding tokens, rather than address ordering, dominated the tested setup.

INDUSTRY EXPERIENCE

Zhipu AI | Multimodal Algorithm Research Intern *Oct 2025 - Feb 2026*
GLM-4.6V Series | Beijing, China

- Contributed to multimodal alignment and long-video understanding for GLM-4.6V, using slime and SGLang in its reinforcement-learning training pipeline.
- Built large-scale multimodal data pipelines for cleaning, filtering, and quality control.

ADDITIONAL TECHNICAL EXPERIENCE

Tsinghua University Supercomputing Team | Trainee *2026 - Present*

- Training in parallel programming, GPU/HPC optimization, and performance profiling alongside the team.

Tsinghua University | Quicksviewer Multimodal LLM *Feb 2025 - Aug 2025*
Academic Rising Stars Program | Department of Computer Science

- Ran multimodal data preprocessing, vision-encoder alignment, and momentum-based chunking experiments for visual question answering and fine-grained long-video understanding.

Open-Cowork | Open-Source Desktop AI Agent | GitHub *2026*

- Built Electron/React/TypeScript features and maintained Skills, document-generation, and WeChat integration workflows across Windows and macOS.

TECHNICAL SKILLS

Programming: Python, C/C++, CUDA, Triton, TypeScript
ML Training: PyTorch, slime, SGLang
GPU Systems: DeepGEMM, FlashInfer, FlashMLA, FLA, Nsight Compute/Systems, Kineto/CUPTI

SELECTED HONORS

2024 Ranked 1st in Gansu Province, National College Entrance Examination
2023 National Bronze Medal, 40th Chinese Physics Olympiad Final
2025 Outstanding Student Leader, Tsinghua University
2024 Freshman First-Class Scholarship, Tsinghua University